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ARTIFICIAL INTELLIGENCE IN FINANCIAL RISK

RiskPulse: NLP Sentiment Engine & Strategic Portfolio Stress Testing Platform

Candidate: Abhi Pandey (Reg No: 21BCE10462)

Department: Computer Science & Engineering (AI & Machine Learning)

Institution: VIT Bhopal University, Madhya Pradesh

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Live URL: <https://vit-bhopal-abhi-pandey-s-p-crisil-h.vercel.app/>

RISKPULSE

INTRODUCTION TO RISKPULSE

What It Is: An autonomous risk intelligence platform bridging unstructured textual financial data with quantitative portfolio stress-testing.

Key Components: Real-time RSS feed ingestion, FinBERT natural language sentiment scoring, and beta-weighted macroeconomic scenario simulation.

Value Proposition: Enables asset managers to detect and mitigate drawdowns before prices drop.

THE FINANCIAL RISK PARADIGM

Traditional Risk Modeling: Relies exclusively on backward-looking historical prices (Historical VaR) and overnight batch processing. By the time market prices drop, portfolios have already absorbed catastrophic losses.

80%+ Unstructured Text: Over 80% of market-moving intelligence is textual—breaking news, earnings calls, central bank statements, and regulatory filings. Modern desks require proactive natural language parsing to bridge qualitative headlines directly into balance-sheet stress tests.

CORE CHALLENGES IN FINANCIAL RISK

- 1. Latency-Vulnerability Gap:** Macroeconomic developments take hours or days to be priced into legacy risk models, exposing capital to severe drawdowns.
- 2. Financial Lexicon Misclassification:** Generic NLP models misread domain jargon (e.g., 'liability shrink' or 'hawkish pause'), producing dangerous false positives.
- 3. Siloed Stress-Testing:** Portfolio stress testing exists as static spreadsheet exercises isolated from streaming market news signals.
- 4. Multi-Sector Contagion:** Single-stock market shocks trigger unmonitored ripple effects across correlated suppliers and competitors.

PROJECT MOTIVATION

Capital Preservation: Sudden macroeconomic shifts require instantaneous portfolio hedging rather than lagging end-of-day reviews.

Academic Innovation: Synthesizing domain-specific FinBERT transformers with rigorous econometric Value-at-Risk (VaR) formulas.

Democratizing Tech: Providing real-time risk intelligence without \$25k/yr proprietary terminal licenses.

CORE OBJECTIVES

- 1. Live Ingestion:** Continuous RSS feed extraction with resilient mock fallback.
- 2. FinBERT NLP:** Financial polarity scoring (-1.0 to +1.0) and impact severity (1-10).
- 3. Entity NER:** Sector-aware extraction mapping company mentions to stock tickers.
- 4. Macro Stress Engine:** Dynamic scenario modeling (Rate Hikes, Stagflation) with Beta repricing.
- 5. Tail Risk (VaR):** Parametric Value-at-Risk and Expected Shortfall recalculation.
- 6. Cloud Verification:** Reactive React 18 dashboard with 100% test pass rate (30/30 tests).

EXISTING SYSTEMS & SHORTCOMINGS

Bloomberg / Refinitiv Terminals: Extreme annual subscription costs exceeding \$25,000 per seat; relies on manual analyst screening and closed proprietary code.

Legacy Statistical VaR (RiskMetrics): Exclusively backward-looking over 250 or 500 trading days; completely blind to breaking news catalysts until prices drop.

Generic NLP Analyzers (VADER): Trained on social media; misclassifies financial jargon and lacks entity mapping.

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PROPOSED SYSTEM CONCEPT FLOW

Autonomous Pipeline: [USER] → [RAW INPUTS] → [INGESTION & NLP] → [STRESS MODULES] → [REAL-TIME TELEMETRY].

Provides transparent, deterministic econometric modeling without generative hallucinations, ensuring regulatory alignment and sub-second execution.

SYSTEM ARCHITECTURE

Tier 1 - Ingestion: Async RSS feed polling, MD5 hash deduplication, synthetic fallback stream.

Tier 2 - FinBERT Core: Deep learning sentiment (-1..+1), impact rating (1..10), ticker NER.

Tier 3 - Stress Engine: Macro shocks, beta scaling, VaR/CVaR recalculation, React 18 UI.

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END-TO-END SYSTEM WORKFLOW

Stage 1: Data Polling via async RSS workers.

Stage 2: Text Normalization and hash deduplication.

Stage 3: FinBERT forward inference for polarity S in $[-1.0, +1.0]$.

Stage 4: Impact severity heuristic (1-10) and ticker NER resolution.

Stage 5: Portfolio stress propagation via Beta-weighted shock formulas.

Stage 6: Live UI refresh displaying updated valuations and risk gauges.

METHODOLOGY & FORMULATIONS

- **Sentiment Polarity:** $S = P(\text{Positive}) - P(\text{Negative}) \in [-1.0, +1.0]$
- **Impact Severity:** $I = \text{round}(1 + 9 \cdot |S| \cdot C_{\text{event}}) \in [1, 10]$
- **Asset Shock:** $\Delta P_i = \text{Base_Shock} \cdot \beta_i \cdot (1 + I_i / 10)$
- **Tail Risk (VaR):** $\text{VaR}_\alpha = V_p \cdot z_\alpha \cdot \sigma_p$; $\text{CVaR}_\alpha = V_p \cdot [\phi(z_\alpha) / (1 - \alpha)] \cdot \sigma_p$

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TECHNOLOGY STACK

Frontend: React 18, Vite, Tailwind CSS, Lucide Icons, Axios.

Backend: Python 3.11+, FastAPI (ASGI), Uvicorn, Pydantic v2.

AI / ML: FinBERT (ProsusAI), PyTorch, Hugging Face Transformers.

Database & Feeds: SQLite 3, SQLAlchemy ORM, Feedparser, Requests.

Testing & Cloud: Pytest (30 test suites), HTTPX, Vercel, Render.

CORE SYSTEM MODULES

- **Ingestion Module (backend/ingestion.py):** Async RSS feeds and synthetic streaming fallback.
- **NLP Engine (backend/nlp_engine.py):** FinBERT transformer scoring and ticker NER.
- **Stress Engine (backend/portfolio.py):** Macro shock simulation, Beta multipliers, and VaR.
- **Web Dashboard (frontend/src/):** Reactive user interface with real-time risk gauges.

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IMPLEMENTATION HIGHLIGHTS

Non-Blocking Ingestion: FastAPI lifespan handlers run background async workers for RSS feeds, ensuring zero latency degradation on user queries.

Dual-Engine NLP Fallback: Full FinBERT PyTorch inference with seamless rule-based CPU failover preventing 500 errors.

Strict Pydantic Contracts: Guarantees rigid schema validation across all REST boundaries.

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APPLICATION INTERFACE

Panel 1 (Signal Stream & Gauge): Streaming ingested headlines, sentiment chips, and market polarity meter.

Panel 2 (Stress Panel & Heatmap): Interactive scenario shock triggers, asset repricing, and VaR telemetry.

KEY PLATFORM FEATURES

- 1. Real-Time Ingestion:** Continuous news extraction with automated deduplication.
- 2. FinBERT Sentiment:** Domain-specific polarity scoring (-1.0 to +1.0).
- 3. Impact Severity:** Algorithmic 1-10 severity scale.
- 4. Sector NER:** Direct ticker extraction and mapping.
- 5. Macro Stress Testing:** Real-time scenario simulation.
- 6. Tail Risk VaR / CVaR:** Quantitative downside loss modeling.

EXPERIMENTAL RESULTS

- **REST API Latency:** < 35 ms average response time across all endpoints.
- **Ingestion Throughput:** 42.8 documents/second processed and scored on single CPU core.
- **FinBERT Inference:** 31.8 ms per headline evaluation.
- **Stress Simulation:** 1.4 ms computation time for a 5-asset portfolio shock.
- **Test Suite:** 30 passed unit and integration tests (100% pass rate).

ADVANTAGES & LIMITATIONS

Core Advantages: Zero LLM hallucinations (deterministic math); sub-35ms ASGI execution; Basel III alignment; accessible open architecture.

Current Limitations: Static beta assumptions (no GARCH); single-node SQLite database; dictionary-based NER heuristics.

FUTURE TRENDS & ROADMAP

Milestone 1 (1-3 Mo): TimescaleDB migration, GARCH(1,1) volatility, 25+ global feeds.

Milestone 2 (3-6 Mo): Causal knowledge graphs for contagion, ONNX INT8 quantization (<10ms).

Milestone 3 (6-12 Mo): Live broker API order execution, multi-agent conversational risk copilot.

CONCLUSION & REFERENCES

Conclusion: Successfully demonstrates that modern lightweight NLP and asynchronous web frameworks can deliver institutional-grade financial risk intelligence.

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